

Optimal Continuity Allocation for a Tightly-coupled KF-based GNSS/IMU Navigation System with Redundant IMUs

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ABSTRACT

This paper introduces an optimal continuity allocation algorithm for a tightly-coupled Kalman-filter (KF)-based Global Navigation Satellite System (GNSS) and multiple Inertial Measurement Units (IMUs) integrated navigation system. Beyond our recent work on the integrity and continuity algorithm of a KF-based GNSS and a single IMU navigation system, the method proposed in this paper enables IMU sensor exclusion while assuring total continuity and integrity requirements under redundant IMU sensor configuration. Two filtering scenarios in the presence of redundant IMU sensors are considered: 1) individual filters that utilize state estimates from a single set of filters at a time, and uses other filters as backup solutions, and 2) a decentralized filter where all the parallel local filters process each IMU sensor measurement and local estimates are subsequently fused in a master filter to achieve the global estimations. First, an analytical equation that can evaluate the continuity risk probability under each filter scenario is derived. Second, based on the fact that sub-filters within the decentralized filter share the same GNSS measurements with different IMU sensors, an inter-filter correlation between test statistics of the decentralized filter is considered to tightly allocate continuity to each monitor. Inter-filter correlation is quantified recursively by the KF, and the monitor thresholds are determined from the formulated joint distributions by the inter-filter correlation. Lastly, a minimum bounding protection level (PL) is determined by optimally allocating continuity risk to each monitor. Continuity allocation considering the correlation significantly improves the overall availability by lowering the continuity burden of each monitor. Simulation results show the benefits of utilizing redundant IMU in terms of system availability and the benefits of taking into account the exact inter-filter correlation for the case of a decentralized filter when allocating continuity requirements for each monitor.

1. INTRODUCTION

Recently, autonomous vehicles, including autonomous drones or self-driving cars, are getting a great deal of attention with the expectation that they can partially replace or augment human performance while improving operational efficiency and increasing the range of applications. Since those vehicles conduct control and path-following operations autonomously by trusting the given navigation information, the availability of autonomous vehicles is predicated on a sufficient level of navigation requirement.

Recent studies suggest that the required level of the navigation requirement of autonomous vehicles should be set comparable to that of mandated in civil aviation [1]. In response, researchers leverage the concept of navigation algorithms developed for civil aviation when developing autonomous vehicle navigation systems.

One key research transition when developing the navigation system of autonomous vehicles compared to that of civil aviation is to consider various multiple navigation sensors and different kinds of filtering algorithms. Currently, multiple sensors, including the Global Navigation Satellite System (GNSS), Inertial Measurement Unit (IMU), vision sensor, barometer, magnetometer, and LIDAR, are integrated to provide the navigation solution of autonomous vehicles. As the many inputs from multiple sensors are integrated, navigation solutions become more vulnerable to potential sensor faults. Potential integrity and continuity risks from all the sensors used for the system should be considered along with its integration algorithm to assure the integrity and continuity of a multi-sensor navigation system.

While there could be various combinations of sensor integration scenarios, a tightly-coupled Kalman filter (KF)-based GNSS/IMU navigation system is generally utilized as a basis for typical multi-sensor navigation systems. Since the KF is a recursive estimator, a recursive sensor fault impact on KF state estimates should be considered when quantifying integrity risks. The KF based integrity algorithms that quantify the recursive fault impact on state estimates are actively being discussed in current publications [2-4]. These previous studies dealt with sensor faults that occur in the GNSS measurement sequence, which is applied for the KF measurement

update, assuming the IMU is a fault-free sensor.

However, studies reported that the IMU is also vulnerable to potential faults, and the reliability of an inertial sensor is critically vital for operations [5]. When GNSS and IMU are integrated with a KF, IMU sensor biases are generally augmented in a KF state vector and are continuously estimated from the filter based on an error dynamics model. In a closed-loop KF algorithm, the estimated IMU biases for each axis are fed back and subtracted from an input IMU measurement at the next epoch. However, this bias estimation process is not enough for capturing IMU faults because: 1) IMU faults may not follow the pre-defined error dynamics model, and 2) even if we assume that IMU faults follow the pre-defined error model, there is a time delay to perfectly capture the faults. Thus, the integrity algorithm for the IMU faults occurring in the state prediction step is also required in multi-sensor navigation systems to fully assure the system safety, based on the fact that there are two main filtering steps in KF, which are a state prediction and a measurement update.

In our previous work [6], we illustrated how unexpected step type IMU faults (potentially due to fracture, stiction, or delamination) affect the KF estimates when they are used for the state prediction in a GNSS/IMU integrated navigation system. It was also shown that the fault impact on KF state estimates due to the IMU sensor faults could be expressed using the KF innovation vector in real-time, whereas this is not applicable in the case of GNSS sensor faults.

In our following work [7], we proposed two protection levels (PLs) determined for the IMU sensor fault hypothesis using the KF innovation vector and additional noise bounding terms. One is a “real-time $PL_{KF,IMU}$ ” that supports position error bounding in real-time by accounting for actual fault and noise impact on user position, and the other is a “predictive $PL_{KF,IMU}$ ” that can be used for availability evaluation before initiating the operation based on the determined worst-case position error after fault monitoring. Based on the two PL concepts as well as PLs for nominal condition and GNSS fault hypothesis, the operation concept was suggested both prior to operation and in a real-time operation against the IMU sensor fault hypothesis.

Through the studies in [6] and [7], we established an integrity and continuity architecture of tightly-coupled GNSS/IMU navigation systems against all sensor faults, including the IMU sensor fault hypothesis. However, the architecture was preferentially designed by assuming that only a single IMU is utilized for the system without redundancy. Thus, a single IMU sensor failure was assumed to guarantee the integrity and continuity risk probabilities assigned to the IMU failure hypothesis when designing an integrity and continuity fault trees, a fault monitor, and a PL equation. The critical limitation of not having redundant IMU sensors is that sensor exclusion is meaningless in the case of IMU sensor fault because no further sensor utilization is possible, and navigation integrity of the integrated navigation system is not guaranteed after IMU exclusion.

Currently, most autonomous navigation systems utilize multiple redundant IMU sensors to improve robustness. There have been many studies on fault detection and isolation algorithms in the presence of redundant IMU sensors. However, algorithms that quantify the continuity and integrity risk considering the recursive IMU fault impact on user position within the KF in the presence of redundant IMUs have not been investigated. Thus, we must apply the developed integrity and continuity algorithms to realistic operational conditions where GNSS-integrated redundant IMU sensors are available.

Beyond our recent work in [6] and [7], this paper builds an extended concept of the continuity and integrity algorithms for a tightly-coupled KF-based GNSS/IMU integrated navigation system by accounting for multiple redundant IMU sensors with a special focus on continuity allocation under multiple IMU fault hypotheses. The integrity and continuity architecture is capable of IMU sensor fault detection and exclusion while assuring total system navigation requirements. Two filter design scenarios in the presence of multiple redundant IMU sensors [8] are considered:

- (a) an individual filter that utilizes state estimates from a single set of filters at a time, and uses other filters as backup solutions,
- (b) a decentralized filter where all the parallel local filters process each IMU sensor measurement and local estimates are subsequently fused in a master filter to achieve the global estimations.

Integrity and continuity algorithms for the individual filter is a simple extension of the previously developed algorithm with additional redundant parallel filters so that modification on the risk allocations, fault detection, and PL computation algorithms are not necessary. However, we can take advantage of parallel filters to ensure integrity in the event of an IMU sensor failure without availability interruption. On the contrary, integrity and continuity algorithms for the decentralized filter should be modified so that the filter meets the total navigation requirements against all fault hypotheses of the multiple IMU sensors. In particular, we have to consider correlation between sub-filters that share same GNSS measurements. A previous work has investigated the impact of inter-filter correlation of a decentralized filter on system continuity and integrity by changing an unknown inter-filter correlation ranges from 0 to 1 for a multi-sensor system [9]. In our study, we analytically quantified the inter-filter correlations between sub-filter test statistics sharing the same GNSS measurements but using different IMU sensors to tightly allocate continuity to each monitor. The quantified inter-filter correlation is applied when evaluating the total continuity risk of a decentralized filter using a total continuity equation of the decentralized filter derived in this paper. Note that this study optimally allocates only the probability of false-alert

continuity risk to monitors under fault-free conditions, and assumes that a portion of continuity risk is separately pre-allocated to true fault conditions. The continuity risk in the event of a failure may vary depending on IMU sensor quality. Thus, further analysis is required for each IMU sensor used for potential applications and it will be addressed in future studies. Lastly, we determined a minimum bounding PL by optimally allocating continuity risk to each monitor.

2. IMU SENSOR FAULT IMPACT ON A KALMAN FILTER BASED GNSS/IMU NAVIGATION SYSTEM

2.1. Tightly-coupled KF-based GNSS/IMU navigation system

The tightly-coupled KF based GNSS/IMU navigation system estimates errors in user attitude ($\delta\phi_{xyz}$), velocity (δv_{xyz}), and position (δp_{xyz}). In addition to those navigation states, biases included an accelerometer ($b_{a_{xyz}}$), and gyroscope measurements ($b_{g_{xyz}}$) are estimated. Lastly, GNSS receiver clock bias (clk) and clock bias drift rate ($clkr$) are estimated and used to formulate the predicted pseudorange using IMU predicted states. The state vector of the GNSS/IMU system is shown in (1) [10, 11].

$$\delta x_k = \begin{bmatrix} \delta\phi_{xyz} & \delta v_{xyz} & \delta p_{xyz} & b_{a_{xyz}} & b_{g_{xyz}} & clk & clkr \end{bmatrix} \quad (1)$$

Through the KF, the state vector is sequentially estimated. The KF has two main steps, which are a state prediction and a measurement update. Equations for the two EKF steps are shown in (2) and (3), respectively.

$$\delta \bar{x}_k = \Phi_k \delta \hat{x}_{k-1} + \vec{\omega}_k \quad (2)$$

$$\delta \hat{x}_k = \delta \bar{x}_k + K_k (\delta z_k - H_k \delta \bar{x}_k) \quad (3)$$

where,

$\delta \bar{x}_k$ is the predicted state vector at time k

$\delta \hat{x}_k$ is the measurement updated state vector at time k

Φ_k is the state transition matrix, and the function of inertial sensor measurements

$\vec{\omega}_k$ is the process noise vector

K_k is the Kalman gain matrix

δz_k is the error between GNSS measurement and predicted measurement

H_k is the observation matrix

clk is the receiver clock offset

$clkr$ is the receiver clock drift

2.2. Unexpected IMU fault impact on a KF-based GNSS/IMU navigation system

When GNSS and IMU are integrated with a KF, IMU sensor biases are generally augmented in a KF state vector and are continuously estimated from the filter based on an error dynamics model [10]. In a closed-loop KF algorithm, the estimated IMU biases for each axis are fed back and subtracted from an input IMU measurement at the next epoch. However, this bias estimation process is not enough for capturing IMU faults because: 1) IMU faults may not follow the pre-defined error dynamics model, and 2) even if we assume that IMU faults follow the pre-defined error model, there is a time delay to perfectly capture the faults.

This section overviews the unexpected IMU fault impact on a KF-based GNSS/IMU navigation system. The derivation can also be found in our previous related paper [6]. To see the impact of IMU fault on a KF state error, a nominal state vector $\delta \bar{x}_k$ is replaced

with the faulty state vector $\delta \bar{x}_k$ derived in (4). A predicted state at time k becomes,

$$\begin{aligned}\delta \bar{x}_k' &= (\Phi_k + \Delta \Phi_k) \delta \hat{x}_{k-1} \\ &= \Phi_k \delta \hat{x}_{k-1} + \vec{f}_{\delta \bar{x}_k}\end{aligned}\quad (4)$$

The IMU sensor measurements are applied to formulate the state transition matrix Φ_k . If the IMU sensor measurement has a deviation from the true sensor measurement due to sensor faults, it results in a deviation in the state transition matrix, denoted as $\Delta \Phi_k$. We define the resulting IMU sensor fault impact on the predicted state as the fault vector $\vec{f}_{\delta \bar{x}_k}$. The resulting fault vector propagates through the KF recursively. Substituting (4) into (3) yields a measurement updated state as (5).

$$\begin{aligned}\delta \hat{x}_k' &= \delta \bar{x}_k' + K_k' (\delta \bar{z}_k - H_k \delta \bar{x}_k') \\ &= (I - K_k' H_k) (\Phi_k \delta \hat{x}_{k-1} + \vec{f}_{\delta \bar{x}_k}) + K_k' \delta \bar{z}_k\end{aligned}\quad (5)$$

Here, K_k' represents the Kalman gain matrix corrupted by the faulty state transition matrix. For simplicity, we introduce a new matrix $L_k' = (I - K_k' H_k)$. Subsequently, (5) is simplified as (6):

$$\begin{aligned}\delta \hat{x}_k' &= L_k' (\Phi_k \delta \hat{x}_{k-1} + \vec{f}_{\delta \bar{x}_k}) + K_k' \delta \bar{z}_k \\ &= L_k' \Phi_k \delta \hat{x}_{k-1} + K_k' \delta \bar{z}_k + L_k' \vec{f}_{\delta \bar{x}_k}\end{aligned}\quad (6)$$

Based on the derived expression in (6), it is noted that the vector $L_k' \vec{f}_{\delta \bar{x}_k}$ is additionally added due to the IMU fault measurements at time k. Finally, the state error equation is derived using (7), where the true state is expressed as (8).

$$\delta \tilde{x}_k = \delta \hat{x}_k' - \delta \bar{x}_k \quad (7)$$

$$\delta \bar{x}_k = \Phi_k \delta \bar{x}_{k-1} + \vec{\omega}_k \quad (8)$$

The resulting state error equation is shown in (9), and the derivation procedure of the state error is introduced in Appendix A.

$$\begin{aligned}\delta \tilde{x}_k &= L_k' \Phi_k \delta \tilde{x}_{k-1} + K_k' \vec{v}_k - L_k' \vec{\omega}_k + L_k' \vec{f}_{\delta \bar{x}_k} \\ &= L_k' (\Phi_k \delta \tilde{x}_{k-1} + \vec{f}_{\delta \bar{x}_k}) + K_k' \vec{v}_k - L_k' \vec{\omega}_k\end{aligned}\quad (9)$$

A state error at time k can be expressed as a function of a previous state vector, a state fault vector, a measurement noise vector, and a process noise vector. The state error equation indicates that an unexpected IMU fault affects the KF state error. It is necessary to consider IMU sensor faults within the integrity/continuity architecture to fully assure the system navigation requirements. The following section introduces an integrity/continuity risk allocation tree of a KF-based GNSS/IMU navigation system, which includes a redundant IMU sensor fault hypothesis as well as the nominal condition and GNSS failure hypotheses.

2.3. Integrity/Continuity risk allocation tree of a KF-based GNSS/IMU navigation system with redundant IMU sensors

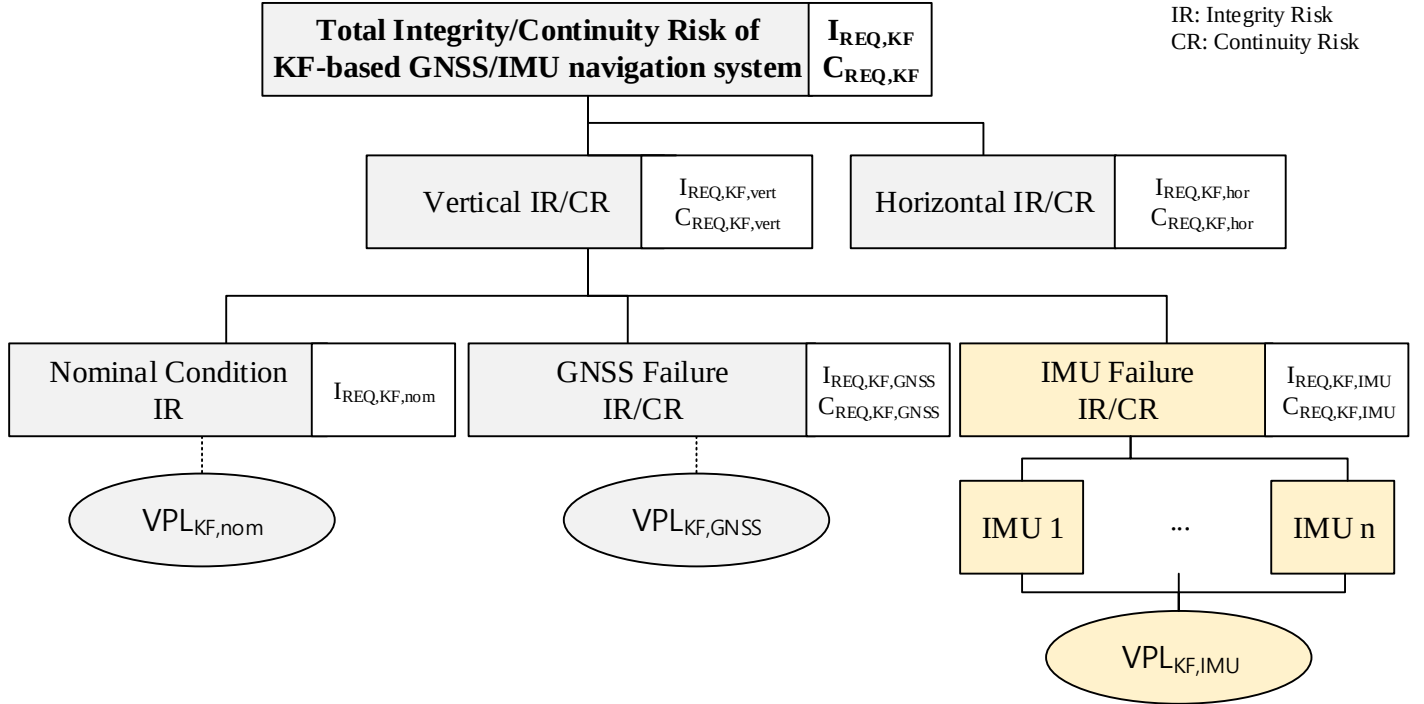


Figure 1 Integrity/continuity risk allocation tree of a KF-based GNSS/IMU navigation system with redundant IMU sensors

Figure 1 introduces an integrity/continuity risk allocation tree of a KF-based GNSS/IMU navigation system with redundant IMU sensors. The fault tree takes into account all possible sensor fault hypotheses to assure the integrity and continuity requirements of a multi-sensor navigation system. The probability of required integrity ($I_{REQ,KF}$) and required continuity ($C_{REQ,KF}$) that a system should meet are separated into the vertical domain ($I_{REQ,KF,vert}$, $C_{REQ,KF,vert}$) and the horizontal domain ($I_{REQ,KF,hor}$, $C_{REQ,KF,hor}$). Figure 1 describes a design for the fault-tree in the vertical position domain as an example. In this study, three fault hypotheses are defined including a nominal hypothesis, a GNSS fault hypothesis, and the IMU sensor faults hypothesis. The total $I_{REQ,KF,vert}$ and $C_{REQ,KF,vert}$ are allocated to each hypothesis, and the $I_{REQ,KF,vert}$, $C_{REQ,KF,vert}$ for each hypothesis are used in the fault monitors as well as the corresponding PL computations with prior probabilities of GNSS and IMU sensors. For each fault hypothesis, PLs are computed. The previous study introduced the PL equations for each fault hypothesis for a GNSS and a single IMU fault hypothesis [7]. In this paper, we focus on the IMU sensor faults when redundant IMUs are integrated to a GNSS with a KF.

3. INTEGRITY/CONTINUITY ALGORITHM OF A KF-BASED GNSS/IMU NAVIGATION SYSTEM WITH REDUNDANT IMUS

This section discusses the integrity and continuity algorithm of a KF-based GNSS/IMU navigation system with redundant IMUs. The algorithms are designed for two different filtering schemes under redundant IMUs (individual filters and a decentralized filter). First, individual filters utilize state estimates from a single set of filters at a time and use other filters as backup solutions. In a decentralized filter, all the parallel local filters process each IMU sensor measurement, and local estimates are subsequently fused in a master filter to achieve the global estimations. Figure 2 shows the concept of two different filtering schemes of GPS/IMU integration using redundant IMUs. Suppose that one has three different IMUs and a single GNSS receiver, one can generate three redundant navigation state estimates by running three local filters for each IMU. For individual filters, state estimates of the main filter are used for navigation unless any of the fault monitors of the main filter is tripped. The entire integrity and continuity risk for the IMU faults are allocated to the main filter. A decentralized filter, on the contrary, integrates local navigation state estimates to

generate a final complete state estimate ($\hat{x}_m$) and state covariance matrix (P_m) in a master filter. A weighted least square (WLS) estimator is a well-known method for fusion of local filter estimates [8]. The integrity and continuity risk should be allocated appropriately to those local filters.

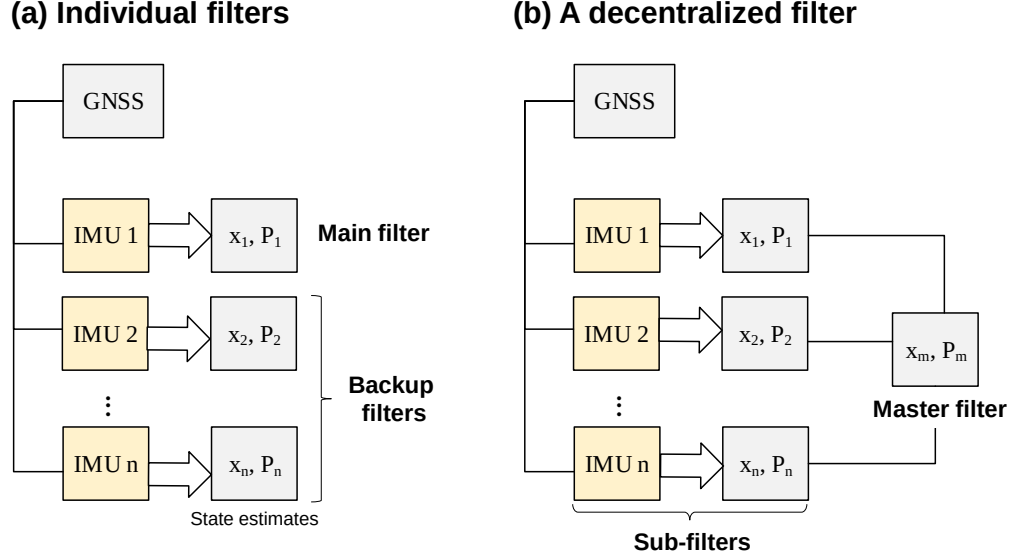


Figure 2. Two different strategies to utilize a single common GNSS receiver and redundant IMUs: (a) individual filters, and (b) a decentralized filter.

3.1. Individual filters using redundant IMUs

3.1.1. Integrity/Continuity probability allocation

Under the individual filter scenario, integrity and continuity algorithm design is a simple extension of the previously proposed algorithm for a GNSS and a single IMU integrated navigation system. The only difference is that we can utilize backup filters when the main filter is flagged as faulty. The integrity and continuity risk allocation to the individual filters is not necessary because the main filter takes the entire amount of integrity and continuity risk. This section briefly reviews the concept of an integrity/continuity algorithm for an individual filter scenario.

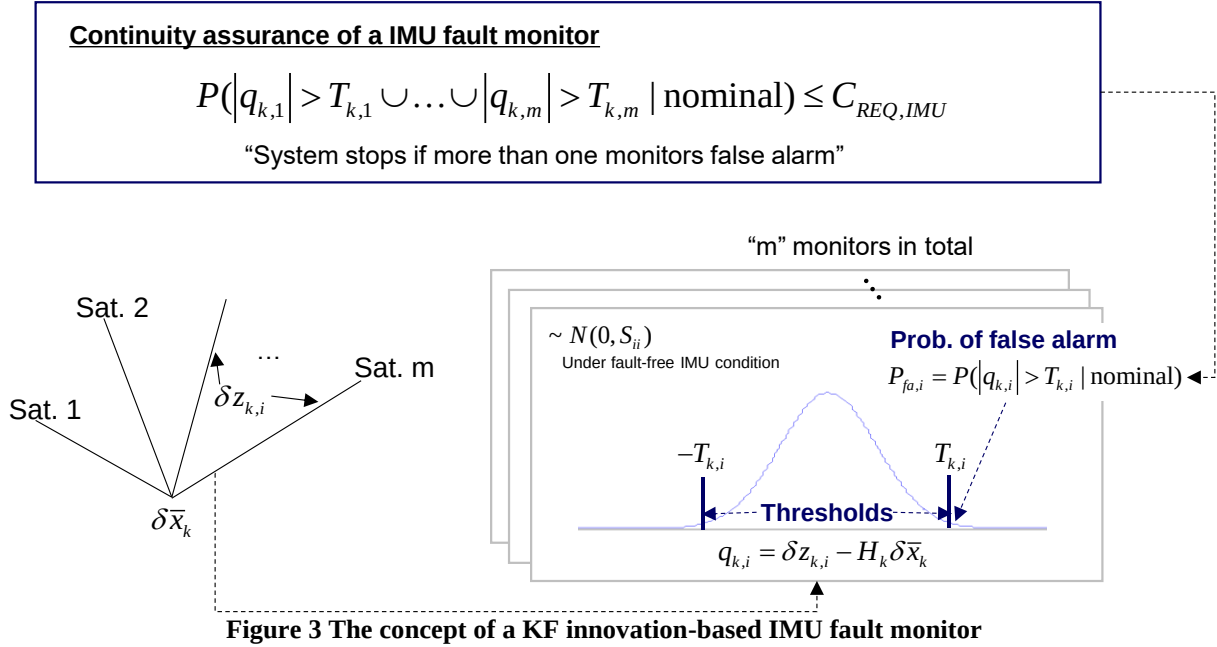
3.1.2. IMU fault detection, and exclusion algorithm

A KF innovation in a GNSS/IMU integrated system is defined as the difference between the GNSS pseudorange measurements and the IMU sensor predicted range. The IMU sensor predicted range is determined using a position estimate computed by IMU sensor measurements and the GNSS satellite position information. Under the IMU sensor fault condition, the predicted KF state is corrupted by the IMU faults. It results in faulty KF innovation vectors. Utilizing this fact, EKF innovation vectors are employed to monitor IMU sensor faults assuming GNSS measurements are fault-free. The GNSS measurement fault is assumed to be monitored in a GNSS fault monitor. (10) represents the KF innovation under IMU sensor faults.

$$\begin{aligned}
 \tilde{\gamma}_k &= \delta \tilde{z}_k - H_k \delta \tilde{x}_k \\
 &= H_k (\Phi_k \delta \tilde{x}_{k-1} + \tilde{\omega}_k) + \tilde{v}_k - H_k (\Phi_k \delta \hat{x}_{k-1} + \tilde{f}_{\delta \tilde{x}_k}) \\
 &= -H_k \Phi_k \delta \tilde{x}_{k-1} + H_k \tilde{\omega}_k + \tilde{v}_k - H_k \tilde{f}_{\delta \tilde{x}_k} \\
 &= -H_k (\Phi_k \delta \tilde{x}_{k-1} + \tilde{f}_{\delta \tilde{x}_k}) + H_k \tilde{\omega}_k + \tilde{v}_k
 \end{aligned} \tag{10}$$

It is observed that a KF innovation and the state error vector include the common fault vector component, $\Phi_k \delta \tilde{x}_{k-1} + \vec{f}_{\delta \tilde{x}_k}$. This finding motivates the utilization of a KF innovation for monitoring and bounding the state error derived in (9).

To monitor IMU sensor faults, a KF innovation monitor is employed. Figure 3 shows the concept of the designed KF innovation monitor for detecting IMU sensor faults. In our monitor design, all the KF innovation vectors formulated using each GNSS measurement are used to monitor the faults, and if more than one monitors false alert, the system stops operation. In this situation, the continuity probability is expressed as the union of all false alert probabilities under nominal conditions. Monitor thresholds are determined to assure the total continuity probability allocated to the IMU fault hypothesis from the fault tree introduced in Section 2. The total number of determined thresholds becomes m .



3.1.3. Protection level (PL) against IMU sensor faults

This section overviews two different PLs, a real-time PL and a predictive PL against IMU sensor faults that were introduced in our recent work [7]. In the previous sections, state error and a KF innovation vector are derived in (9), and (10), respectively. It is noted that the state error can be expressed utilizing the KF innovation vector with additional GNSS noise terms. The state error can be expressed as a function of the KF innovation (11). The subscript p indicates the KF position states.

$$\tilde{x}_{k,p} = -B_{k,p} \vec{\gamma}_k + (B_{k,p} + K_{k,p}') \vec{v}_k \quad (11)$$

where,

$$B_{k,p} = L_{k,p}' (H_{k,p}^T H_{k,p})^{-1} H_{k,p}^T \quad (12)$$

Based on (11), two protection level concepts have been proposed in [7]. (13) shows a real-time PL that utilizes the KF innovation vector to estimate the real-time position bias against IMU sensor faults. Additional bounding terms that account for KF innovation noise and remaining state error noise are applied to bound from the errors up to the required level of system integrity. Since the real-time PL concept does not employ any fault monitor, no continuity probability is assigned when computing the real-time PL.

$$\text{Real-time VPL}_{IMU,KF} = -\vec{t}_v B_{k,p} \vec{\gamma}_k \pm K_{md,IMU} \cdot \sigma_{(B_{k,p} + K_{k,p}') \vec{v}_k, v} \quad (13)$$

(14) shows the predictive $VPL_{KF,IMU}$. The predictive VPL assures the user vertical position error after monitoring the IMU fault using the KF innovation-based fault monitor as introduced in Section 3.1.2. The worst magnitude of undetected faults are converted into the position domain to determine the worst position bias ($b_{worst,k,v}$), and additional noise terms are considered to formulate the predictive $VPL_{KF,IMU}$.

$$\begin{aligned} \text{Predictive } VPL_{KF,IMU} &= \left| -\vec{t}_v B_{k,p} \vec{T}_{worst,k} \right| + K_{md,IMU} \cdot \sigma_{\tilde{x},v} \\ &= b_{worst,k,v} + K_{md,IMU} \cdot \sigma_{\tilde{x},v} \end{aligned} \quad (14)$$

Since the main focus of this paper is to evaluate IMU fault monitor performance when redundant IMUs are employed, the rest of the paper focuses on the predictive PL which takes into account both continuity and integrity risk whereas the real-time PL only applies integrity risk.

3.2. A decentralized filter using redundant IMUs

In a decentralized filter strategy, the EKF innovation-based fault detection monitors are implemented in each local filter. The key difference of a decentralized filter from an individual filter strategy is the inter-filter correlation due to the common GNSS receiver. This causes the EKF innovations, which are used for the IMU fault detection, from different local filters to be correlated as well, and should be taken into account when the continuity and integrity algorithms are designed. The following sub-sections describe the effect of inter-filter correlation on EKF innovations and the determination of the PL.

3.2.1. Correlation between monitor test statistics of local filters

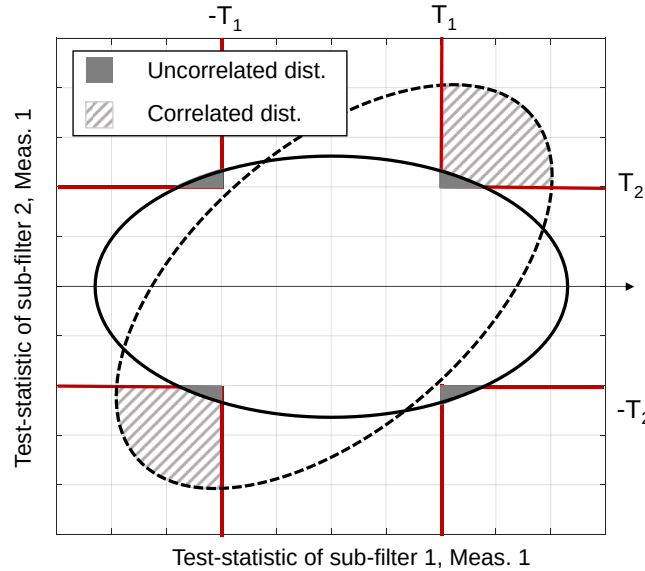


Figure 4 Continuity risk of uncorrelated (shaded area) and correlated (hatched area) test statistic between two local filters

Figure 4 shows a simplified graph that illustrates how the continuity risk of a correlated distribution differs from that of an uncorrelated distribution. The circle with solid line and ellipse with dotted line indicate the same amount of confidence interval of joint probability density function for uncorrelated and correlated distributions, respectively. Red lines indicate thresholds for innovation monitors. Continuity risk can be calculated as the integration of a bivariate normal distribution over the shaded areas. The filled area in gray and the hatched area indicate the continuity risk for the uncorrelated and the correlated distribution. Note that continuity risk can be varied with the correlation between the test statistics of local filters, which in turn affects the risk allocation process.

Since the parallel local filters share the common GNSS receiver, the local state estimates will be correlated to each other. Previous research pointed out that inter-filter correlation should be taken into account because continuity risk increases as inter-filter

correlation increases in a decentralized filter [9]. The innovation vector of the i^{th} local filter at epoch k is denoted as $\gamma_{k,i}$. Suppose that there are two local filters as a simple example, then the innovation vector of the two local filters can be derived as (15)

$$\begin{aligned}\gamma_{k,1} &= -H_{k,1}\Phi_{k,1}\delta\tilde{x}_{k-1,1} + H_{k,1}\bar{w}_{k-1,1} + \nu_k \\ \gamma_{k,2} &= -H_{k,2}\Phi_{k,2}\delta\tilde{x}_{k-1,2} + H_{k,2}\bar{w}_{k-1,2} + \nu_k\end{aligned}\quad (15)$$

The correlation of the innovation vectors can then be derived as

$$E[\gamma_{k,1}\gamma_{k,2}^T] = H_{k,1}\Phi_{k,1}E[\delta\tilde{x}_{k-1,1}\delta\tilde{x}_{k-1,2}^T]\Phi_{k,2}^TH_{k,2}^T + R_k \quad (16)$$

where,

$$\begin{aligned}\delta\tilde{x}_{k,1} &= L_{k,1}\Phi_{k,1}\delta\tilde{x}_{k-1,1} - L_{k,1}\bar{w}_{k-1,1} + K_{k,1}\nu_k \\ \delta\tilde{x}_{k,2} &= L_{k,2}\Phi_{k,2}\delta\tilde{x}_{k-1,2} - L_{k,2}\bar{w}_{k-1,2} + K_{k,2}\nu_k\end{aligned}\quad (17)$$

$$E[\delta\tilde{x}_{k,1}\delta\tilde{x}_{k,2}^T] = L_{k,1}\Phi_{k,1}E[\delta\tilde{x}_{k-1,1}\delta\tilde{x}_{k-1,2}^T]\Phi_{k,2}^TL_{k,2}^T + K_{k,1}R_kK_{k,2}^T \quad (18)$$

According to the above equation (16), the common GNSS measurement noise term R_k is included in the correlation term, which implies that correlation always exists between the innovation vectors of local filters using common GNSS receivers.

3.2.2. Protection level considering the inter-filter correlation

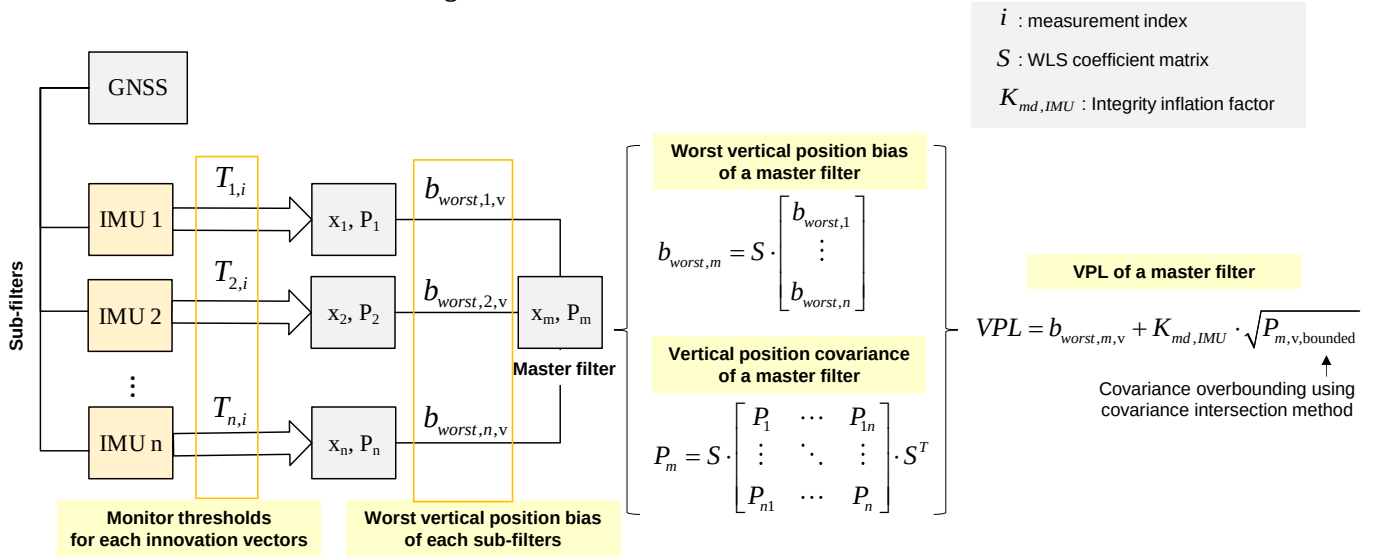


Figure 5 VPL computation algorithm of a decentralized filter

Figure 5 depicts a scheme of the calculation of protection levels in a decentralized filter. From each local filter, thresholds are determined for each KF innovation vector. The determined threshold vectors are used to compute the worst vertical position bias. The worst vertical position bias and state covariance matrix are integrated to generate the parameters of a master filter. Protection level is then calculated as a sum of the worst position bias and the uncertainty bound of a master filter, whose magnitude is determined via the risk allocation (continuity risk for the former term and integrity risk for the latter term). The correlation between the local filters should be taken into account in the continuity and integrity algorithm, which will now be addressed.

3.2.2.1. Continuity risk allocation

As mentioned above, the innovation vector is utilized as an IMU fault detection monitor. Let n be the number of IMUs and m be the number of GNSS measurements at a particular epoch. The total number of operating monitors then becomes mn . Total continuity risk budget should be separately allocated to mn fault monitors. The simplest method of allocation is dividing the total continuity risk budget into the same amount of probability across mn innovation monitors of all the local filters. However, the correlation between the monitors is not accounted for in this method, which can degrade navigation performance.

In a single Kalman filter, any event of false alert for the innovation monitors under the nominal condition interrupts the navigation system resulting in a continuity risk event. In a decentralized filter, however, one can isolate a specific local filter that has at least one innovation monitor trip. In other words, the continuity risk event for the decentralized filter only occurs when the monitors for all local filters trip at the same time. Let $q_{i,j}$ and $T_{i,j}$ be the monitor test statistic and the threshold of the i^{th} local filter on the j^{th} element of the innovation vector, then the continuity risk C_r is defined as below.

$$C_r \equiv P((|q_{1,1}| > T_{1,1} \cup \dots \cup |q_{1,m}| > T_{1,m}) \cap \dots \cap (|q_{n,1}| > T_{n,1} \cup \dots \cup |q_{n,m}| > T_{n,m}) | \text{nominal}) \quad (19)$$

Exact calculation of the above equation requires computationally expensive processes due to the integration of multivariate joint probability density functions. One way to release the computation load is to split the union of multiple sets into the separate intersection of sets as shown in (20). Here, an upper bound $\bar{C}_r$ on continuity risk C_r can be derived as a sum of individual probabilities representing the separate sets.

$$\bar{C}_r \equiv \sum_{i_n=1}^m \dots \sum_{i_1=1}^m P(|q_{1,i_1}| > T_{1,i_1} \cap \dots \cap |q_{n,i_n}| > T_{n,i_n} | \text{nominal}) \quad (20)$$

A brief explanation can be given with a simplified example represented by Venn diagrams. In Figure 6, it is assumed that there are two local filters sharing two GNSS pseudorange measurements, which can be represented as four sets of events. Each set denoted by $FA_{i,j}$ represents a set of false alert events of each innovation monitor. The overlapped area colored with green on the left-hand side represents the continuity risk, and this area is always smaller than the sum of the four shaded areas on the right-hand side. Thus, the inequality that $\bar{C}_r$ is larger than C_r always holds, and the calculation becomes much simpler.

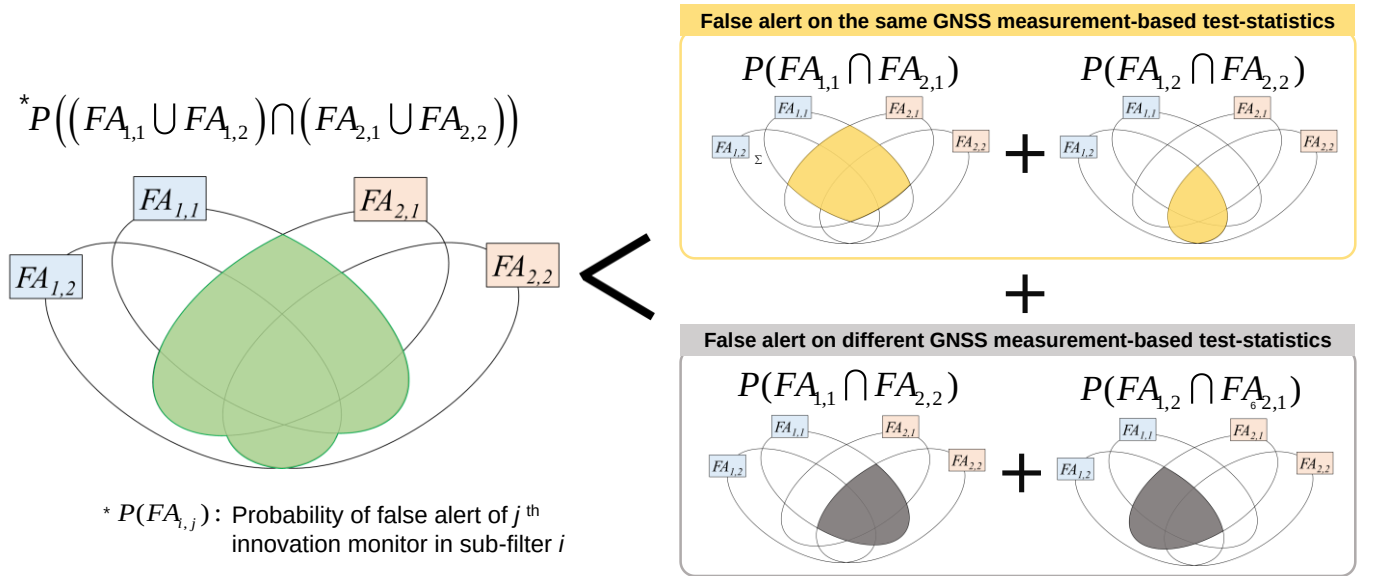


Figure 6 Venn diagram example of the concept of continuity risk bounding

The continuity requirement is fulfilled if the upper bound of continuity risk is less than the required continuity risk $C_{req,KF,IMU}$ as in (21).

$$C_r \leq \bar{C}_r \leq C_{req,KF,IMU} \quad (21)$$

Next, the allocation of the continuity risk can be treated as a non-linear constrained optimization problem of multi variables. The objective function described in (22) is the worst position bias of the master filter output, and optimized variables are the thresholds for the innovation monitors of all the local navigation filters. The final protection level is determined with the minimized worst position bias of the master filter ($b_{worst, master, v}$), so that the problem of minimizing the worst position bias can be formulated as:

$$\min_{T_{1,1} \dots T_{n,m}} b_{worst, master, v} \text{ such that } \bar{C}_r \leq C_{req,KF,IMU} \quad (22)$$

3.2.2.2. Integrity risk allocation

In the master filter, the correlation between the local filters must be accounted for when determining the weighting matrix. The inter-filter correlation should be considered elaborately because poorly modeled or unmodeled correlation can result in an underestimate of the protection level [9]. One can obtain diagonal terms of the weighting matrix from the state covariance matrix of a bank of local filters, but there is no means to determine off-diagonal terms. The matrix overbounding method should be applied to achieve integrity assurance. This can be accomplished by leveraging the covariance intersection algorithm [12, 13]. The covariance intersection algorithm makes all the off-diagonal terms zeroes and searches the optimal scale factors to inflate diagonal terms for compensation of the ignorance of off-diagonal terms.

The uncertainty bound of the protection level is determined by the product of the inflation coefficient $K_{md,IMU}$ and the overbounded covariance. Unlike the continuity risk, however, allocation of the integrity risk is not necessary. The whole integrity risk budget is used to calculate the inflation coefficient. Thus, the inflation coefficient $K_{md,IMU}$ is determined as below

$$K_{md,IMU} = Q^{-1}(I_{req,KF,IMU}) \quad (23)$$

where $Q^{-1}(p)$ is a function that calculates $(1-p)$ quantile of a standard normal distribution.

4. PERFORMANCE EVALUATION

In this section, the developed integrity and continuity algorithms of a KF-based GNSS/redundant IMUs navigation system are evaluated via simulations. Simulations are carried out with a simplified example of individual filters and a decentralized filter that has two sub-filters with two different IMU sensors. Simulation parameters are listed in Table 1.

Table 1 Simulation parameters

Parameter		Value
Requirements	$I_{REQ,KF,IMU}$	0.5×10^{-7}
	$C_{REQ,KF,IMU}$	4×10^{-6}
IMU sensor	Grade	Consumer grade IMU [10]
	Update rate	100 Hz
	The prior probability of sensor fault	10^{-5}
GNSS sensor	Noise level	Standalone GNSS error model [14]
	Update rate	2 Hz
	Constellation	# of satellites: 30, Elevation mask angle: 10 deg.

	The prior probability of sensor fault	10^{-5}
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First, the impact of utilizing redundant IMU sensors in terms of system availability is analyzed by comparing the PL of a single IMU/GNSS system, and dual IMUs/GNSS system. Second, the proposed continuity allocation method by considering inter-filter correlation is evaluated by comparing the PL of a decentralized filter with and without considering inter-filter correlation.

4.1. The benefit of employing redundant IMU sensors in terms of system availability

To simulate the benefit of utilizing redundant IMU sensors on the system availability, the PLs of a single IMU/GNSS system and dual IMUs/GNSS system are compared. We optimally allocated the total continuity requirement of an IMU monitor using (22) and determined thresholds for each innovation monitor. In the case of a single IMU/GNSS filter, total continuity is allocated to innovation monitors within the single filter. When allocating the continuity to each innovation monitor within the single filter, we neglected the correlation between innovation monitors formulated with different GNSS SV measurements. This was done by assuming that inter-satellite correlation is small enough to be ignored and IMU noise common to monitors is much smaller than GNSS measurement noise. In contrast, inter-filter correlation should be additionally considered in the case of the decentralized filter. Under the two sub-filter case, (20) is simplified to (24).

$$\bar{C}_r = \sum_{i_2}^M \sum_{i_1}^M P(|q_{1,i_1}| > T_{1,i_1} \cap |q_{2,i_2}| > T_{2,i_2} | \text{nominal}) \quad (24)$$

Note that there are distinct types of the intersection of sets whether a false alert event occurs at the monitors for the same GNSS measurement or for the different GNSS measurements. In the latter case, the probability of intersection can be approximated to the product of probability of each set since they are nearly uncorrelated with each other. In our simulation, the upper bound $\bar{C}_r$ shown in (24) can be approximated to (25) based on the assumption that correlation between innovation vectors formulated with different GNSS measurements is small enough to be neglected. .

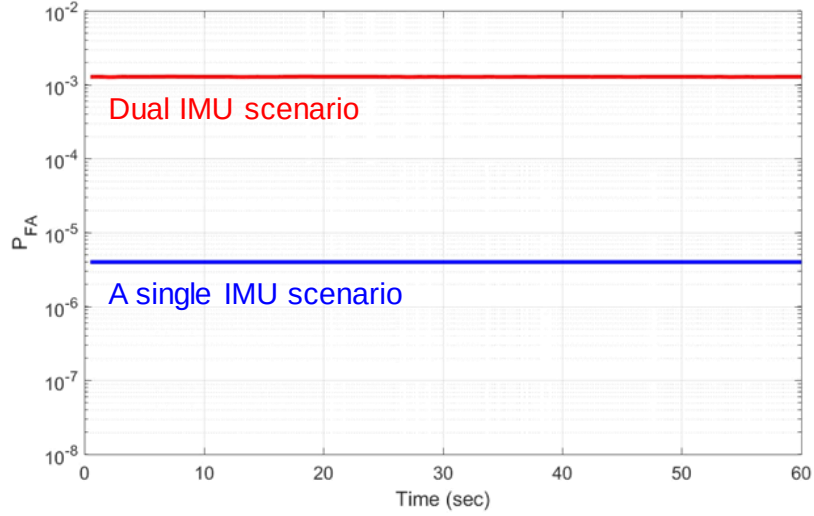
$$\begin{aligned} & \sum_{i_2}^M \sum_{i_1}^M P(|q_{1,i_1}| > T_{1,i_1} \cap |q_{2,i_2}| > T_{2,i_2} | \text{nominal}) \\ & \approx \sum_i^M P(|q_{1,i}| > T_{1,i} \cap |q_{2,i}| > T_{2,i} | \text{nominal}) + \sum_{i_1}^M \sum_{i_2 (i_2 \neq i_1)}^M P(|q_{1,i_1}| > T_{1,i_1} | \text{nominal}) P(|q_{2,i_2}| > T_{2,i_2} | \text{nominal}) \end{aligned} \quad (25)$$

In addition, we considered additional constraints as introduced in (26). This constraint restricts the optimization function from distributing too much continuity probability to one filter. If the total continuity is allocated only to one filter, the rest of the filters will have very low continuity allocation and become nearly useless. This situation may result in lower availability due to the reduced number of filters available. In our simulation, we applied a 10% margin on continuity allocation between two sub-filters.

$$\left| P(FA_{\text{filter\#1}}) - P(FA_{\text{filter\#2}}) \right| \leq P(FA_{\text{filter\#N}}) / 10 \quad (26)$$

Figure 7 shows the allocated continuity probability (Figure 7(a)) and corresponding VPL (Figure 7(b)). The red line indicates the case of utilizing dual IMUs and the blue line shows the case of utilizing only a single IMU. The simulation results show the benefits of employing more than one IMU sensor in terms of availability. Figure 7(a) indicates that a much larger probability of continuity can be allocated to a filter when utilizing dual IMUs as compared to the case of utilizing only a single IMU to meet the same total continuity requirement. The reduced continuity risk allocation ultimately determines lower thresholds and lower VPL represented by solid lines in Figure 7(b). In the dual IMU scenario, lower VPL is achieved which affects higher system availability.

(a) Prob. of false alert allocated to each filter



(b) VPL

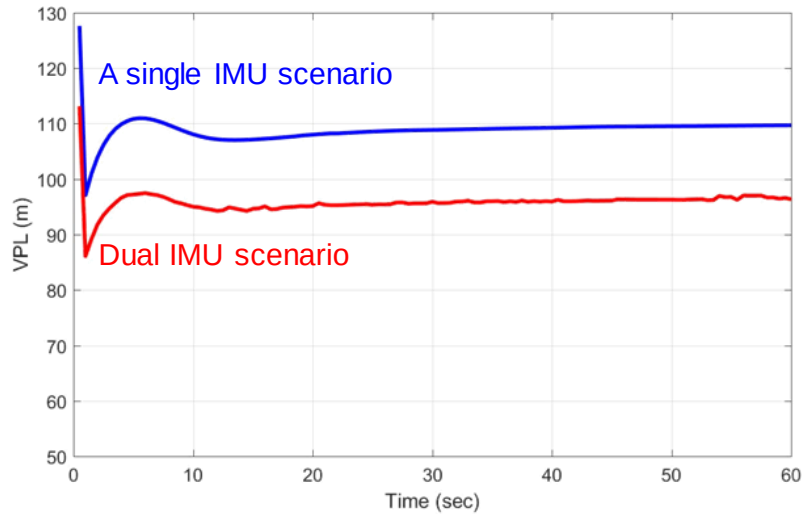


Figure 7 False alert probability allocated for each filter, and corresponding VPL under a single IMU and dual IMUs scenarios

4.2. Tight bounds on the continuity risk by considering the inter-filter correlation

In this section, we see the benefit of system availability when allocating continuity tightly by exploiting information of inter-filter correlation. As discussed in the previous sections, test statistics of KF innovation vectors are correlated between sub-filters because they share the same GNSS measurement noise. Based on the information on inter-filter correlation of the KF innovation vectors, thresholds for each monitor were optimally determined to minimize the worst vertical position bias while meeting the total continuity requirement. The red line in Figure 8(a) shows the continuity allocated to a filter within a decentralized filter considering monitor inter-correlation, and the black line in the same figure shows the continuity allocated to a filter within a decentralized filter under a situation with unknown inter-filter correlation. In the latter case, we need to take a conservative approach, and one of the way to assure system continuity is to allocate the total continuity requirement so that the sum of allocated continuity risks becomes the total system continuity. This can be expressed with (27). (27) bounds the joint probability of the sub-filters's false alert with the sum of all test statistics' false alert probabilities. This approach is conservative by not exploiting correlation information between innovation

vectors under the correlated condition. Conservative continuity risk allocation forces higher continuity burden for each monitor, resulting in lowered availability.

$$\sum_i^N \sum_j^M P(|q_{i,j}| > T_{i,j} \mid \text{nominal}) \leq C_{REQ,KF,IMU} \quad (27)$$

The results in Figure 8(b) indicate that we can meet the continuity requirement more tightly while lowering the final VPL if we exactly consider the inter-filter correlation of the decentralized filter. This analysis illustrates the potential of the proposed algorithm to exploit the inter-filter correlation information to establish tight bounds on the continuity risk.

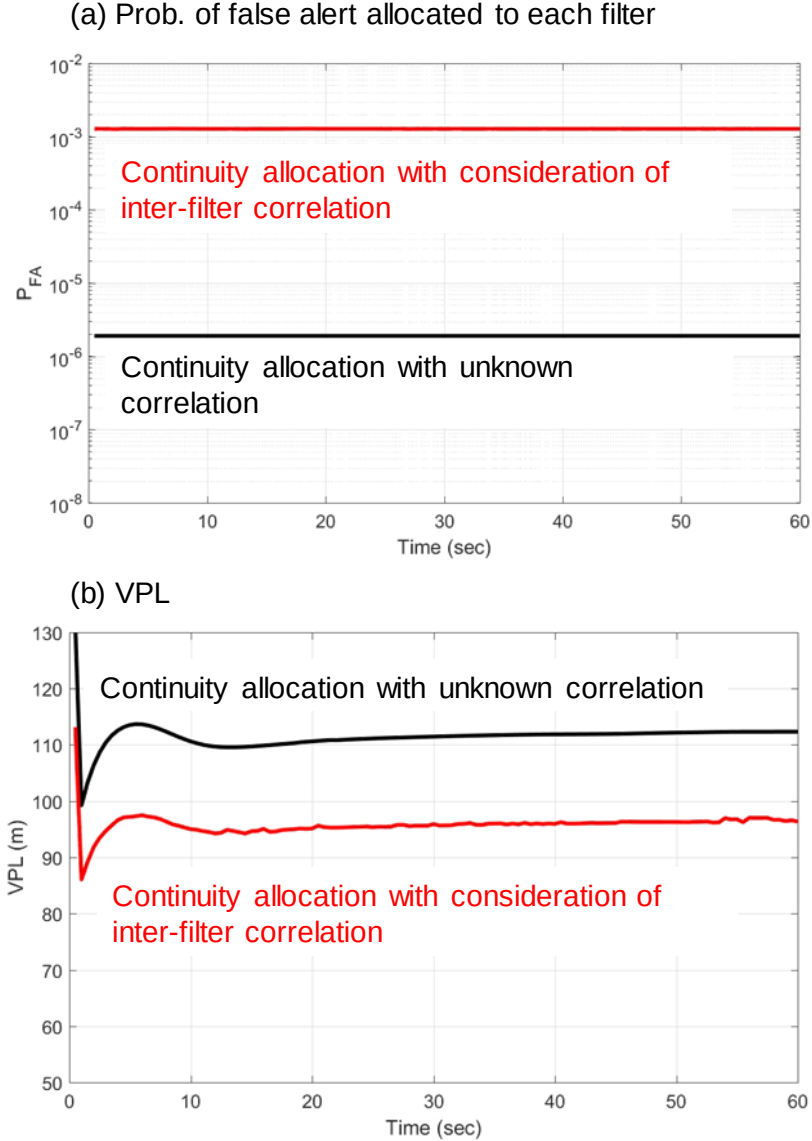


Figure 8 False alert probability allocated for each filter with and without considering monitor inter-correlation, and corresponding VPLs

5. CONCLUSION

We proposed the integrity and continuity algorithms with a special focus on optimal continuity allocation for a tightly-coupled KF based GNSS/redundant IMUs navigation system. In this paper, we mainly focused on continuity risk under a nominal condition, which is allocated to monitor false alert probability when designing a fault monitor. The continuity risk under fault condition should be allocated separately by considering IMU sensor fault probabilities used for each application. The latter part will be further analyzed in future work.

The developed algorithms enable us to quantify the impact of redundant IMUs on system navigation performance in terms of integrity, continuity, and availability. In particular, a decentralized filtering scenario that has multiple sub-filters is considered as an integration filter of multiple IMUs with a GNSS. Continuity allocation to each monitor of sub-filters is newly considered, whereas integrity is assured by bounding the master filter error statistics added to the undetected bias.

For the optimal continuity allocation to each monitor of sub-filters, we first derived an equation that evaluates the total continuity risk of the decentralized filter. By the continuity risk probability bounding method suggested in this paper, the continuity risk can be bounded as a sum of intersections of each monitor false alert probability. Since the sub-filter test statistics are highly correlated with each other, we quantified inter-filter correlation by deriving the analytical equation for the correlation to evaluate the intersections of each monitor false alert probability by formulating joint distributions between monitor test statistics. Lastly, a minimum bounding PL is determined by optimally allocating continuity to each monitor. The optimal allocation to each monitor is conducted by solving an optimization problem in a way to minimize the resulting PL to improve system availability while assuring total continuity as well as integrity.

Performance evaluation is conducted via simulations. Simulation results showed that utilizing the redundant IMUs relaxes the continuity requirements that should be allocated to each sub-filter compared to a single IMU filter case. In addition, if we exploit information on the inter-filter correlation between test statistics of each monitor, we can take benefits from the monitor continuity allocation which reduces the resulting thresholds. The reduced thresholds lower the PL and can improve system availability. The integrity and continuity algorithms proposed in this paper can be broadly applied to the evaluation of navigation system performance of a safety-critical system using a redundant navigation sensor system. Autonomous vehicles are just one example of a system that can benefit from these algorithms.

In this paper, the system integrity is only assured by bounding biased error statistics to the integrity requirement when determining the predictive PL which is used to predict system availability prior to operations. To determine PL that is utilized in real-time operation, integrity risk should be considered for each fault monitor to assure the monitor missed detection probability in addition to the nominal noise bounding. The results of this study showed the addition of redundant IMU sensors to a system relaxes the continuity burden of a monitor and lowers the PL. However, the addition of IMU sensors may tighten the missed detection probability that should be assured in each monitor, resulting in increased PL. The system availability by optimally allocating integrity in addition to continuity under multiple IMU configurations will be further investigated as future work.

APPENDIX A

The steps for deriving the state error equation are as follows,

- 1) Multiply L_k' to $\delta\tilde{x}_k$;

$$L_k' \delta\tilde{x}_k = L_k' \Phi_k \delta\tilde{x}_{k-1} + L_k' \bar{w}_k \quad (28)$$

- 2) Subtract $L_k' \delta\tilde{x}_k$ from $\delta\hat{x}_k'$;

$$\delta\hat{x}_k' - L_k' \delta\tilde{x}_k = L_k' \Phi_k \delta\tilde{x}_{k-1} + K_k \delta\tilde{z}_k - L_k' \bar{w}_k + L_k' \bar{f}_{\tilde{x}_k} \quad (29)$$

- 3) Insert $\delta\tilde{z}_k = H_k \delta\tilde{x}_k + \bar{v}_k$ to (29) where $\bar{v}_k$ is a measurement noise vector;

$$\delta\hat{x}_k' - \delta\tilde{x}_k + K_k' H_k \delta\tilde{x}_k = L_k' \Phi_k \delta\tilde{x}_{k-1} + K_k' (H_k \delta\tilde{x}_k + \bar{v}_k) - L_k' \bar{w}_k + L_k' \bar{f}_{\tilde{x}_k} \quad (30)$$

- 4) Rearrange (30)

$$\delta \tilde{\mathbf{x}}_k = \mathbf{L}_k' \Phi_k \delta \tilde{\mathbf{x}}_{k-1} + \mathbf{K}_k' \tilde{\mathbf{v}}_k - \mathbf{L}_k' \tilde{\boldsymbol{\omega}}_k + \mathbf{L}_k' \tilde{\mathbf{f}}_{\tilde{\mathbf{x}}_k} \quad (31)$$

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